



Material Test Report

11700 GLOGER, HOUSTON, TX 77039
Phone: 800-847-5713 / 281-986-5100 / Fax: 281-986-5151

MTR # : 1322905

Page : 1 of 1

SOLD TO: BIG RED FASTENERS
P.O. BOX 470547
TULSA, OK 74147

SHIP TO: BIG RED FASTENERS
489 NORTH POPLAR AVENUE
RECEIVING
BROKEN ARROW, OK 74012

DATE	SALES ORDER #	CUST P.O.#	TAG NUMBER	SPECIFICATION	
01/05/2026	1748122	B-00059140		ASTM A193 B7, ASTM A320 L7	
LINE #	QTY	ITEM #	ITEM DESCRIPTION	LOT NUMBER	MILL HEAT
1	40		2-8 X 15 ASTM A320 L7 ALL THREAD STUDS XYLAN 1424 BLUE COATED- USA	CSJV	58051250/03

CHEMICAL ANALYSIS

C	Mn	P	S	Si	Cr	Mo
.40	.93	.01	.021	.27	.97	.202

PHYSICAL PROPERTIES

Elongation %	Reduction of Area %	Tensile PSI	Yield PSI	Hardness Brinell	Charpy @ -150 F Ft.Lbf	Charpy @ -150 F Ft.Lbf	MIN. TEMPERING:
22.4	59	139930	120460	293	23	23	1100 F.

SUPPLEMENTAL INFORMATION

ASTM E381: S1/R1/C1 ASTM A193-20 ASTM A320-18 QUENCHED AND TEMPERED
MANUFACTURED, SAMPLED, TESTED, AND INSPECTED IN ACCORDANCE WITH THE
REQUIREMENTS OF THE REFERENCED SPECIFICATION
***COUNTRY OF ORIGIN USA---

Stacy Kellogg
Quality Assurance Department

Job: 74769
PO: 7476910
BOM: 36
Pages: 3
Heat: 58051250/03
C162218
A77769



HIGHLAND THREADS

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DATE	SALES ORDER #	CUST P.O.#	TAG NUMBER	SPECIFICATION	
01/05/2026	1748122	B-00059140		ASTM A194 GRADE 7/7L HVY HEX NUTS 2-8	
LINE #	QTY	ITEM #	ITEM DESCRIPTION	LOT NUMBER	MILL HEAT
5	80		2-8 ASTM A194 GRADE 7L HEAVY HEX NUTS Xylan 1424 BLUE COATED- USA	CYAW	C162218

CHEMICAL ANALYSIS

C	Mn	P	S	Si	Cr	Mo
.41	.85	.017	.014	.25	1.00	.21

PHYSICAL PROPERTIES

Hardness Brinell	24 HR Hard BHN	MIN. TEMPERING:
302	235	1100 F.

SUPPLEMENTAL INFORMATION

ASTM E381: S1/R1/C1 ASTM A194-23 QUENCHED AND TEMPERED CROSS SECTIONAL
HARDNESS: 31.6/31.4/31.8 HRC 32.1/31.9/31.8 HRC CHARPY @ -160 DEG F: 22/20/20 FT/LBS
MANUFACTURED, SAMPLED, TESTED, AND INSPECTED IN ACCORDANCE WITH THE
REQUIREMENTS OF THE REFERENCED SPECIFICATION

***COUNTRY OF ORIGIN USA---

Stacy Kellars

Quality Assurance Department



HIGHLAND THREADS

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DATE	SALES ORDER #	CUST P.O.#	TAG NUMBER	SPECIFICATION	
01/05/2026	1748122	B-00059140		ASTM F436 HARDENED FLAT WASHERS	
LINE #	QTY	ITEM #	ITEM DESCRIPTION	LOT NUMBER	MILL HEAT
6	80		2 F436 FLATWASHERS Xylan 1424 BLUE COATED- USA	CLWA	A77769

CHEMICAL ANALYSIS

C	Mn	P	S	Si
.48	.74	.01	.003	.19

PHYSICAL PROPERTIES

Hardness RC	
43.6	

SUPPLEMENTAL INFORMATION

ASTM F436-11
MANUFACTURED, SAMPLED, TESTED, AND INSPECTED IN ACCORDANCE WITH THE
REQUIREMENTS OF THE REFERENCED SPECIFICATION
100% MELT & MFG IN THE USA

Stacy Kellogg
Quality Assurance Department